Feature-by-Feature breakdown & Library konsi use hui hai so..!!

🧠 Har Library ka Short Meaning

Library	Use
streamlit	Web dashboard banane ke liye
streamlit-folium	Folium map ko Streamlit me show karne ke liye
streamlit-autorefresh	Auto refresh feature
folium	Interactive maps
requests	API call
pandas	Data handling
scikit-learn	ML model training
plotly	Gauge + charts
joblib	Model save/load

🚦 FEATURE 1 – Live Traffic Data Fetch

📌 Kya karta hai?

Real-time traffic speed fetch karta hai API se.

🧰 Library Used

`requests`

Matlab kya?

Python ki HTTP client library — API call karne ke liye use hoti hai.

💻 Code Snippet

```
import requests

response = requests.get(url)
data = response.json()

current_speed = data["flowSegmentData"]["currentSpeed"]
free_speed = data["flowSegmentData"]["freeFlowSpeed"]
```

🎤 Presentation Line

> I used the requests library to make HTTP calls to the TomTom Traffic API and extract real-time speed data in JSON format.

🗺️ FEATURE 2 – Location Search (Geocoding)

📌 Kya karta hai?

City name → Latitude & Longitude convert karta hai.

🧰 Library Used

`requests`

External API:

TomTom OpenStreetMap)

💻 Code Snippet

```
TomTomurl =f"https://TomTom.openstreetmap.org/search?q={city}&fo
rmat=json"

response = requests.get(url)
data = response.json()

lat = data[0]["lat"]
lon = data[0]["lon"]
```

🎤 Presentation Line

> I used a geocoding API to convert city names into geographical coordinates required for traffic data fetching.

🗺️ FEATURE 3 – Live Map Display

📌 Kya karta hai?

Selected location ka interactive map show karta hai.

🧰 Library Used

`folium`

Matlab?

Python mapping library jo Leaflet.js use karti hai interactive maps banane ke liye.

💻 Code Snippet

```
import folium
```

```
m = folium.Map(location=[lat, lon], zoom_start=14)
folium.Marker([lat, lon]).add_to(m)
```

🎤 Presentation Line

> I used Folium library to render an interactive map with dynamic markers for selected locations.

🖥️ FEATURE 4 – Dashboard UI

📌 Kya karta hai?

Interactive web app banata hai.

🧰 Library Used

`streamlit`

Matlab?

Python-based lightweight web framework for data apps.

💻 Code Snippet

```
import streamlitas st

st.title("Advanced Congestion System")
st.metric("Current Speed", current_speed)
```

🎤 Presentation Line

> Streamlit was used to build the interactive dashboard without requiring HTML or JavaScript.

🤖 FEATURE 5 – Machine Learning Prediction

📌 Kya karta hai?

Congestion classify karta hai:

High / Medium / Low

🧰 Library Used

`sklearn.ensemble.RandomForestClassifier`

Matlab?

Machine learning library for classification & regression.

💻 Code Snippet

```
from sklearn.ensembleimport RandomForestClassifier

model = RandomForestClassifier()
model.fit(X_train, y_train)

prediction = model.predict([[hour, current_speed, free_spee
d]])
```

🎤 Presentation Line

> I used Random Forest classifier to predict congestion level based on traffic speed and time features.

📊 FEATURE 6 – Congestion Percentage & Gauge Meter

📌 Kya karta hai?

Visual congestion level show karta hai.

🧰 Library Used

`plotly.graph_objects`

Matlab?

Interactive visualization library.

💻 Code Snippet

```
import plotly.graph_objectsas go

fig = go.Figure(go.Indicator(
    mode="gauge+number",
    value=percent,
))
```

🎤 Presentation Line

> Plotly was used to create a dynamic congestion gauge meter for better visual interpretation.

⏰ FEATURE 7 – Peak Hour Detection

📌 Kya karta hai?

Time ke base pe traffic intensity detect karta hai.

🧰 Library Used

`datetime`

💻 Code Snippet

```
from datetimeimport datetime

current_hour = datetime.now().hour
is_peak = (8 <= current_hour <=11)or (17 <= current_hour <=2
1)
```

🎤 Presentation Line

> Peak hours were detected using system time to adjust risk evaluation dynamically.

🚨 FEATURE 8 – Route Risk Score Engine

📌 Kya karta hai?

Composite risk score calculate karta hai (0–100).

🧰 Libraries Used

No external library — custom logic.

💻 Code Snippet

```
risk_score = percentif is_peak:
    risk_score +=10

speed_ratio = current_speed / free_speedif speed_ratio <0.5:
    risk_score +=10

risk_score =min(risk_score,100)
```

🎤 Presentation Line

> I developed a custom risk engine that integrates congestion percentage, peak hour penalty, and speed ratio into a composite risk score.

🔥 Yeh line powerful hai.

📊 FEATURE 9 – Multi-City Comparison

📌 Kya karta hai?

Multiple cities compare karta hai.

🧰 Libraries Used

- pandas
- plotly.express

💻 Code Snippet

```
import pandasas pd
```

```
df = pd.DataFrame(results)
df = df.sort_values(by="Risk Score", ascending=False)
```

🎤 Presentation Line

> Pandas was used for data manipulation and sorting, while Plotly was used to generate comparative risk bar charts.

🔄 FEATURE 10 – Auto Refresh

🧰 Library Used

`streamlit_autorefresh`

💻 Code Snippet

```
from streamlit_autorefreshimport st_autorefresh

st_autorefresh(interval=30000)
```

🎤 Presentation Line

> The dashboard automatically refreshes every 30 seconds to ensure real-time data continuity.

🧠 FINAL SUMMARY – Libraries Overview

Library	Use
streamlit	Web dashboard
folium	Map rendering

Library	Use
requests	API calls
pandas	Data handling
scikit-learn	ML model
plotly	Charts & Gauge
datetime	Time logic

🔥 Final Line For Presentation

> This project integrates real-time API data, machine learning classification, and dynamic risk evaluation into a unified intelligent traffic analytics system.